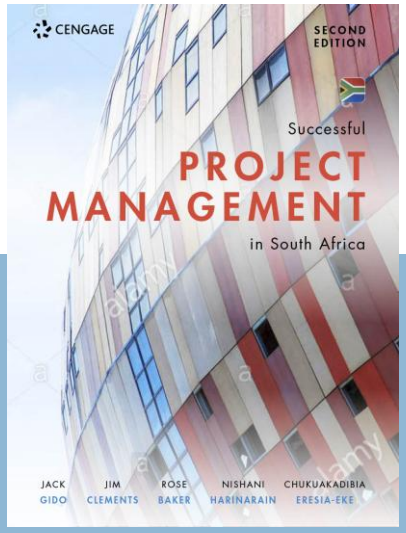


Learning Unit 6

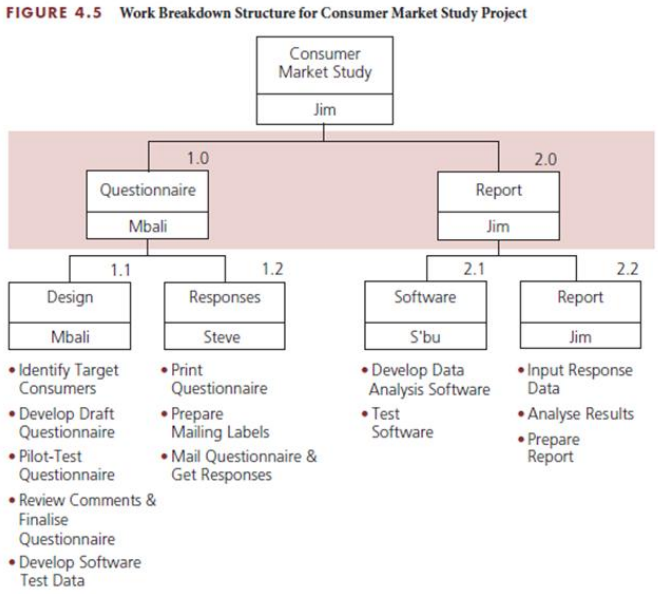
Developing the Schedule and Project Utilisation

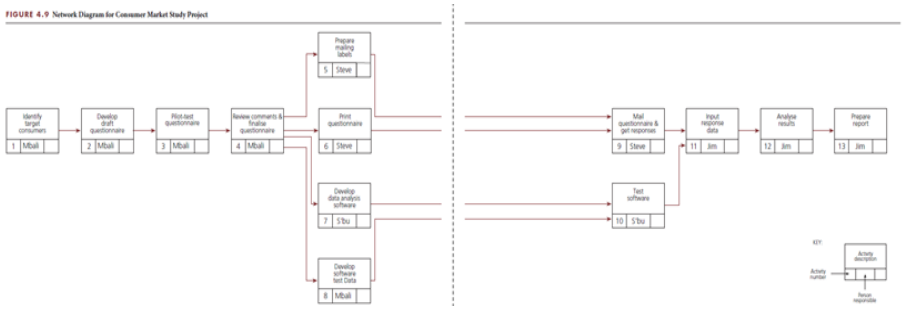
(Chapter 5 and 6)

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Recap…

- Chapter 4 discussed what work needs to be done in terms of scope and deliverables
- It also discussed how that work will get done by defining specific activities and arranging them in a sequence of dependent relationships to create a network plan
- This network plan is a roadmap for how all the activities fit together to achieve the project work scope and objective
- Lastly, we discussed the final phase of the project life cycle is closing the project
- An important project closing action is staff recognition and evaluation





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CHAPTER 5

Developing the Schedule

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Chapter Concepts

- Estimating the resources required for each activity
- Estimating the duration for each activity
- Establishing the estimated start time and required completion time for the overall project
- Calculating the earliest times at which each activity can start and finish, based on the project estimated start time
- Calculating the latest times by which each activity must start and finish in order to complete the project by its required completion time
- Determining the amount of positive or negative slack between the time each activity can start or finish and the time it must start or finish
- Identifying the critical (longest) path of activities
- Performing the steps in the project control process
- Determining the effects of actual schedule performance on the project schedule
- Incorporating changes into the schedule
- Developing an updated project schedule
- Determining approaches to controlling the project schedule
- Implementing agile project management

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Learning Outcomes

- Estimate the resources required for activities
- Estimate the duration for an activity
- Determine the earliest start and finish times for activities
- Determine the latest start and finish times for activities
- Explain and determine total slack
- Prepare a project schedule
- Identify and explain the critical path
- Discuss the project control process
- Develop updated schedules based on actual progress and changes
- Discuss and apply approaches to control the project schedule
- Explain agile project management

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Performance Domains of PMBOK® 7th edition covered in this chapter:		Knowledge Areas of PMBOK® 6th edition covered in this chapter:
• Team • Development Approach & Life Cycle • Planning	• Project Work • Measurement • Uncertainty • Delivery	• Project Integration Management • Project Time Management

Project Management Principles relevant to this chapter

Principle 7 (Tailor Based on Context)
Principle 9 (Navigate Complexity)

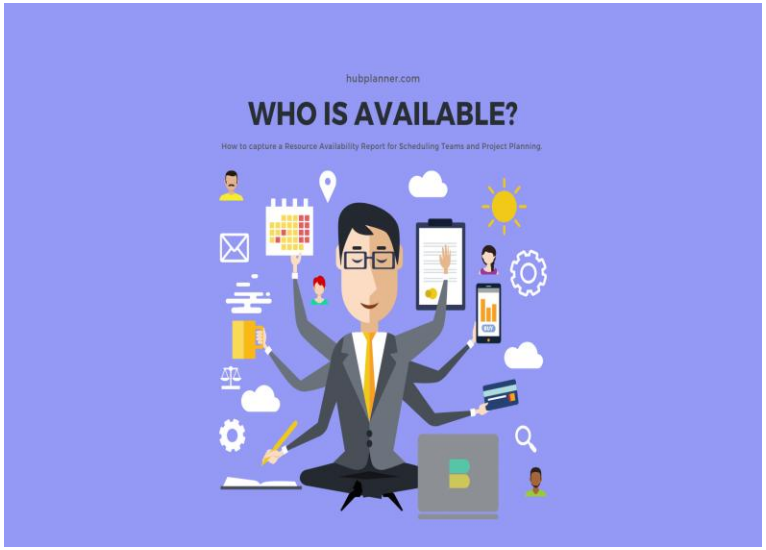
Principle 9 – Complexity: A project can become complex at any point in the project life cycle because of human behaviour, system interactions, uncertainty and ambiguity. Project teams have to be alert to elements of complexity so that they can utilise the appropriate methods to reduce its impact on the project.

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Estimate Activity Resources



- Resources include
  - People, materials, equipment, facilities
- Influence on the duration
  - Availability of the resources
  - Types of resources
  - Sufficient quantities of resources for the activity durations
  - Potential conflicts with other projects may cause
- Involve person with expertise in resource estimate
- Estimates influence costs

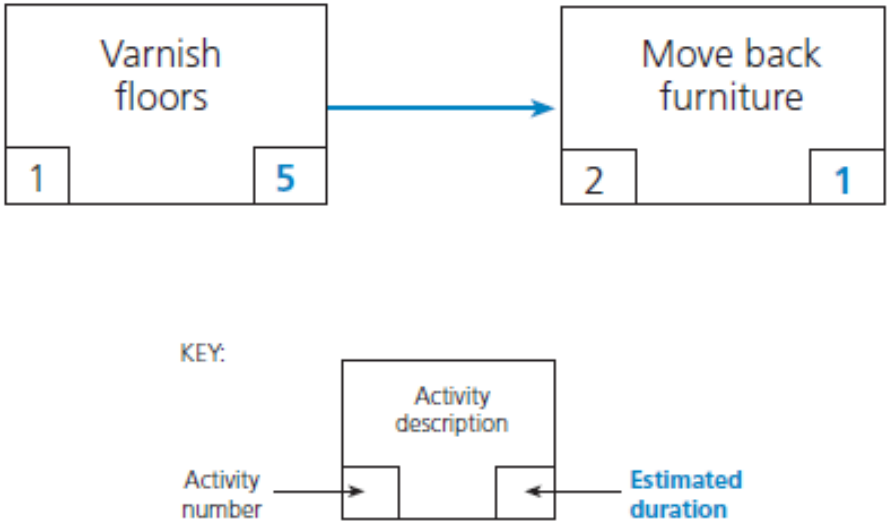


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Estimate Activity Durations

- Duration must be the total elapsed time
  - Time for the work to be done plus any associated waiting time

FIGURE 5.1 Activity Estimated Duration



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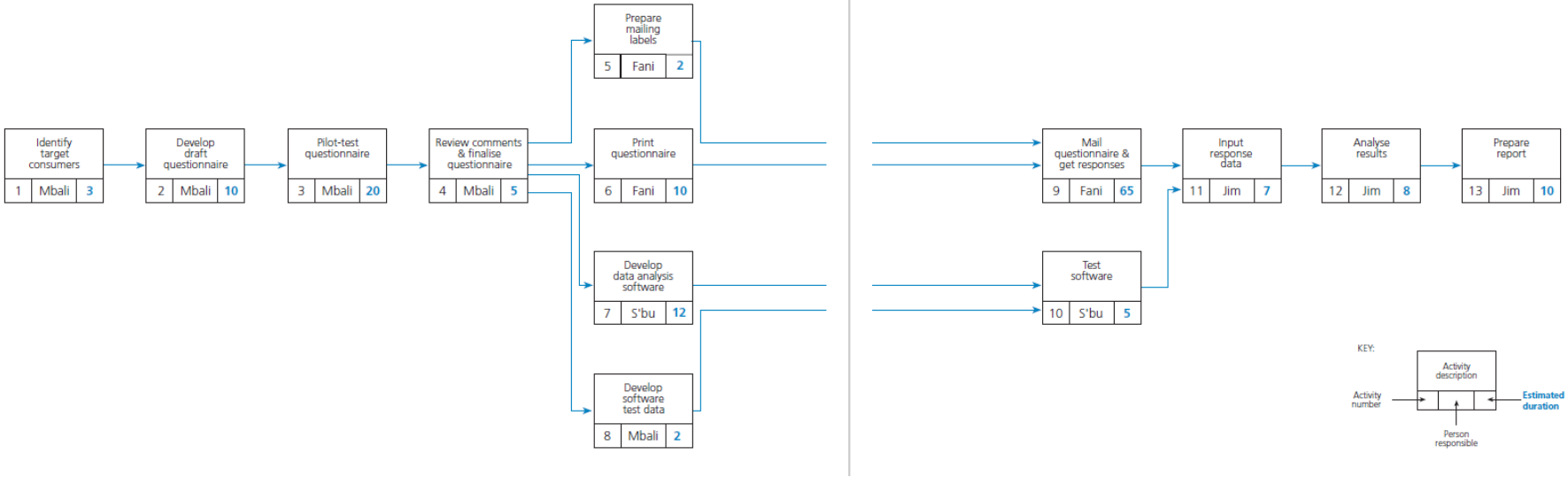
Estimate Activity Durations

- It is a good practice to have the person who will be responsible for performing a specific activity estimate the duration for that activity.
- Designate an experienced individual to estimate the durations for all the activities the organization is responsible in large projects
- Historical data can be a guide for similar or future projects
- Estimated duration should be aggressive yet realistic
- At the beginning of the project, it may not be possible to estimate the durations for all activities with the same accuracy
- The project team can progressively elaborate the estimated durations as more information is known or becomes clear

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Estimate Activity Durations

FIGURE 5.2 Network Diagram for Consumer Market Study Project, Showing Estimated Durations



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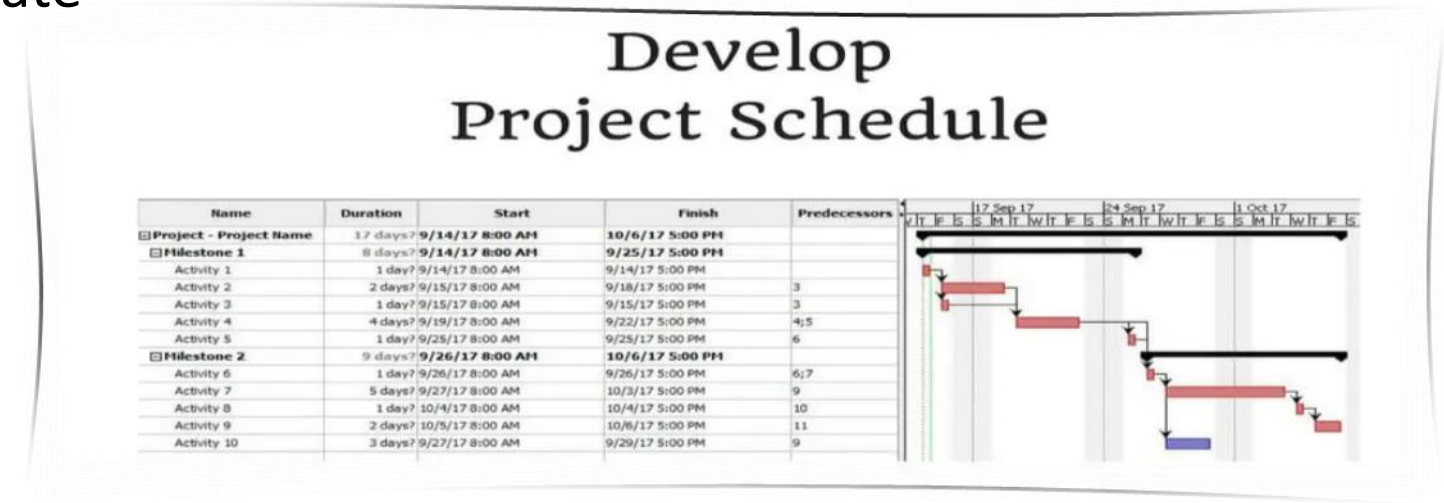
Establish Project Start and Completion Times

- Define the overall window for project completion
- May not want to commit to a specific date
  - Project not to start until customer has approved the contract
  - Delay in contract signing may impact project start
- Set finish time as number of days from project start

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Develop Project Schedule

- Prior activities for schedule development
  - Estimate duration of each activity
  - Establish overall window of time for the project
- Develop the schedule timetable
  - Earliest start and finish times based on estimated start date
  - Latest start and finish times based on required completion date



Name	Duration	Start	Finish	Predecessors
Project - Project Name	17 days?	9/14/17 8:00 AM	10/6/17 5:00 PM	
Milestone 1	8 days?	9/14/17 8:00 AM	9/25/17 5:00 PM	
Activity 1	1 day?	9/14/17 8:00 AM	9/14/17 5:00 PM	
Activity 2	2 days?	9/15/17 8:00 AM	9/18/17 5:00 PM	3
Activity 3	1 day?	9/15/17 8:00 AM	9/15/17 5:00 PM	3
Activity 4	4 days?	9/19/17 8:00 AM	9/22/17 5:00 PM	4;5
Activity 5	1 day?	9/25/17 8:00 AM	9/25/17 5:00 PM	6
Milestone 2	9 days?	9/26/17 8:00 AM	10/6/17 5:00 PM	
Activity 6	1 day?	9/26/17 8:00 AM	9/26/17 5:00 PM	6;7
Activity 7	5 days?	9/27/17 8:00 AM	10/3/17 5:00 PM	9
Activity 8	1 day?	10/4/17 8:00 AM	10/4/17 5:00 PM	10
Activity 9	2 days?	10/5/17 8:00 AM	10/6/17 5:00 PM	11
Activity 10	3 days?	9/27/17 8:00 AM	9/29/17 5:00 PM	9

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A. Earliest start and finish times

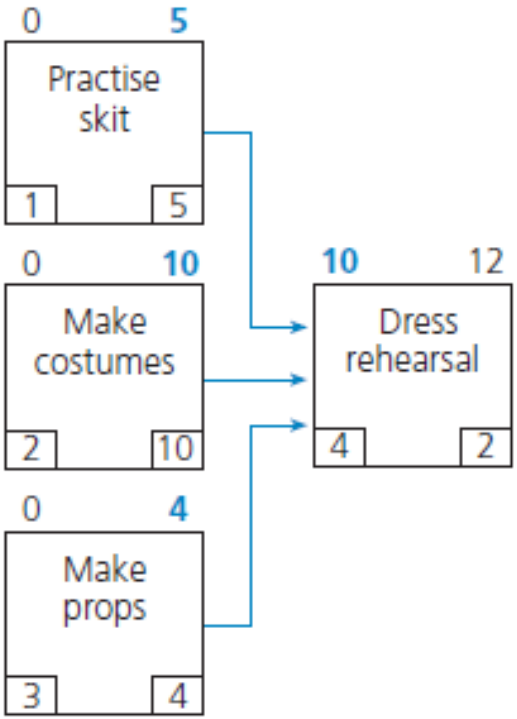
- Earliest start time (ES)
  - Earliest time at which a specific activity can **begin**
- Earliest finish time (EF)
  - Earliest time by which a specific activity can be **completed**
- EF = ES + Estimated duration
- The ES and EF times are determined by calculating ***forward*** through the network diagram → *beginning of the project to the end of the project*
- One rule that must be followed in making these forward calculations
- ***Rule 1:*** *The earliest start time for a specific activity must be the same as or later than the latest of all the earliest finish times of all the activities leading directly into that specific activity*

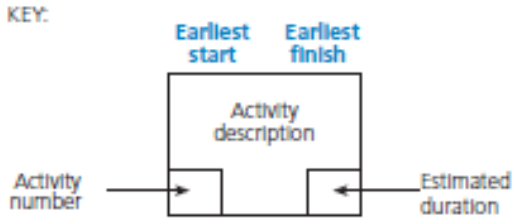
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Earliest start and finish times calculation

- Why is the ES for 'Dress rehearsal’ 10?

FIGURE 5.3 Earliest Start Times



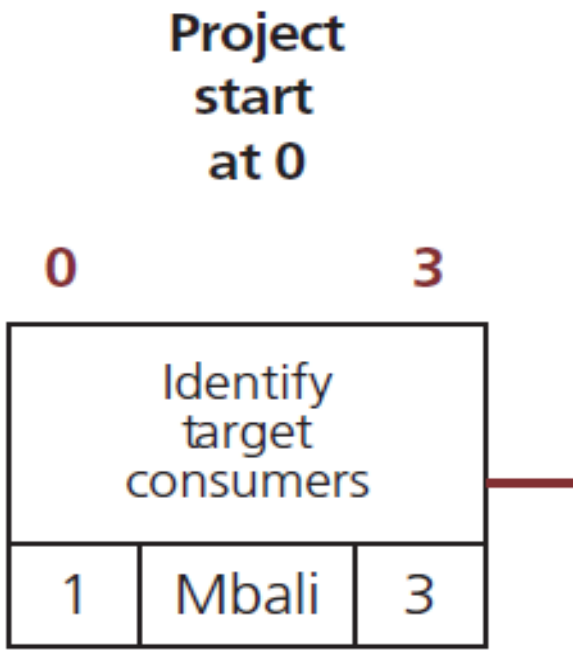


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Figure 5.4 (page 157 – 158): Forward calculations for the consumer market study project

'Identify target consumers'

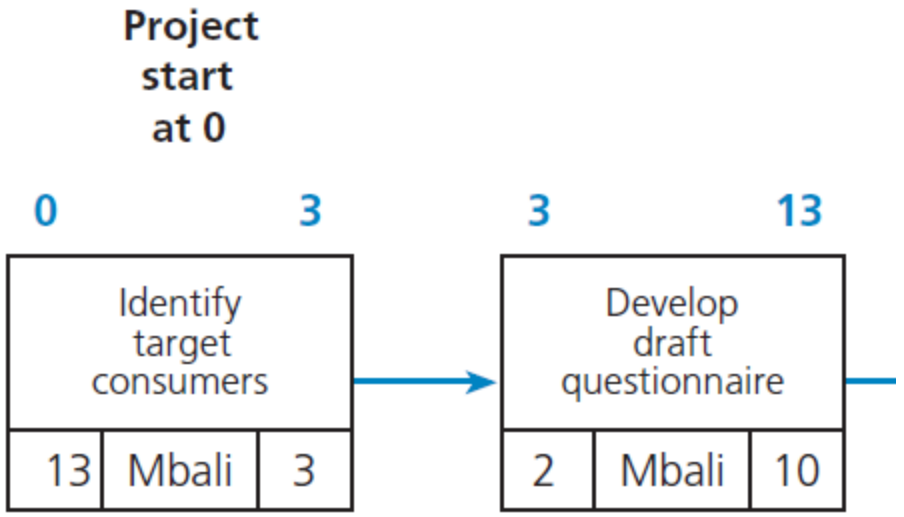
- Start date = 0
- ES = Start date = 0
- Duration = 3
- EF = 0 + 3 = 3



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'Develop draft questionnaire'

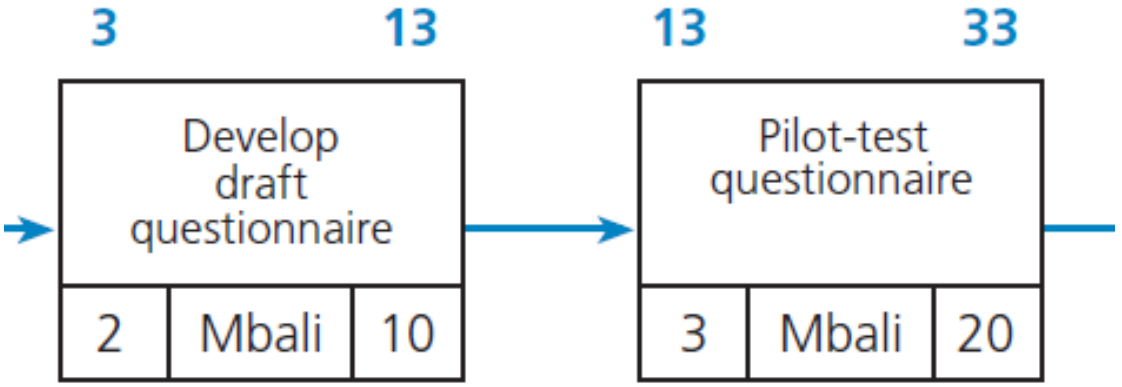
- ES = EF Task 1 = 3
- Duration = 10
- EF = 3 + 10 = 13



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'Pilot-test questionnaire'

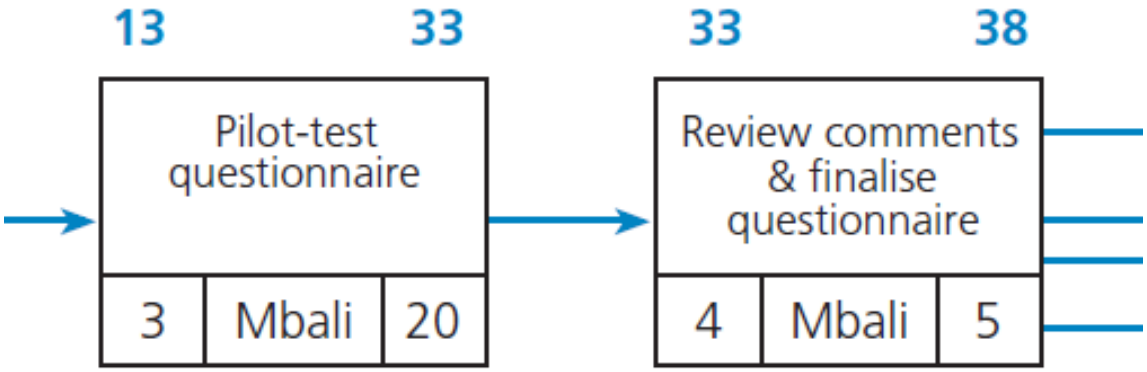
- ES = EF Task 2 = 13
- Duration = 20
- EF = 13 + 20 = 33



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'Review comments & finalise questionnaire'

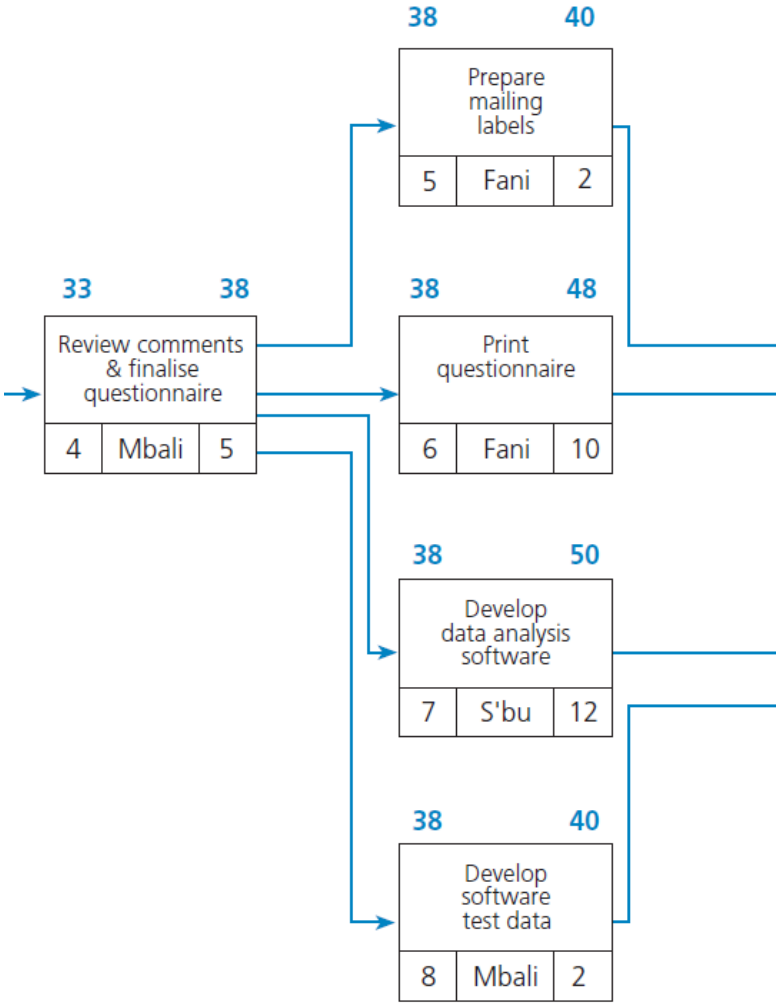
- ES = EF Task 3 = 33
- Duration = 5
- EF = 33 + 5 = 38



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'Prepare mailing labels', 'Print questionnaire', 'Develop data analysis software', and 'Develop software test data'

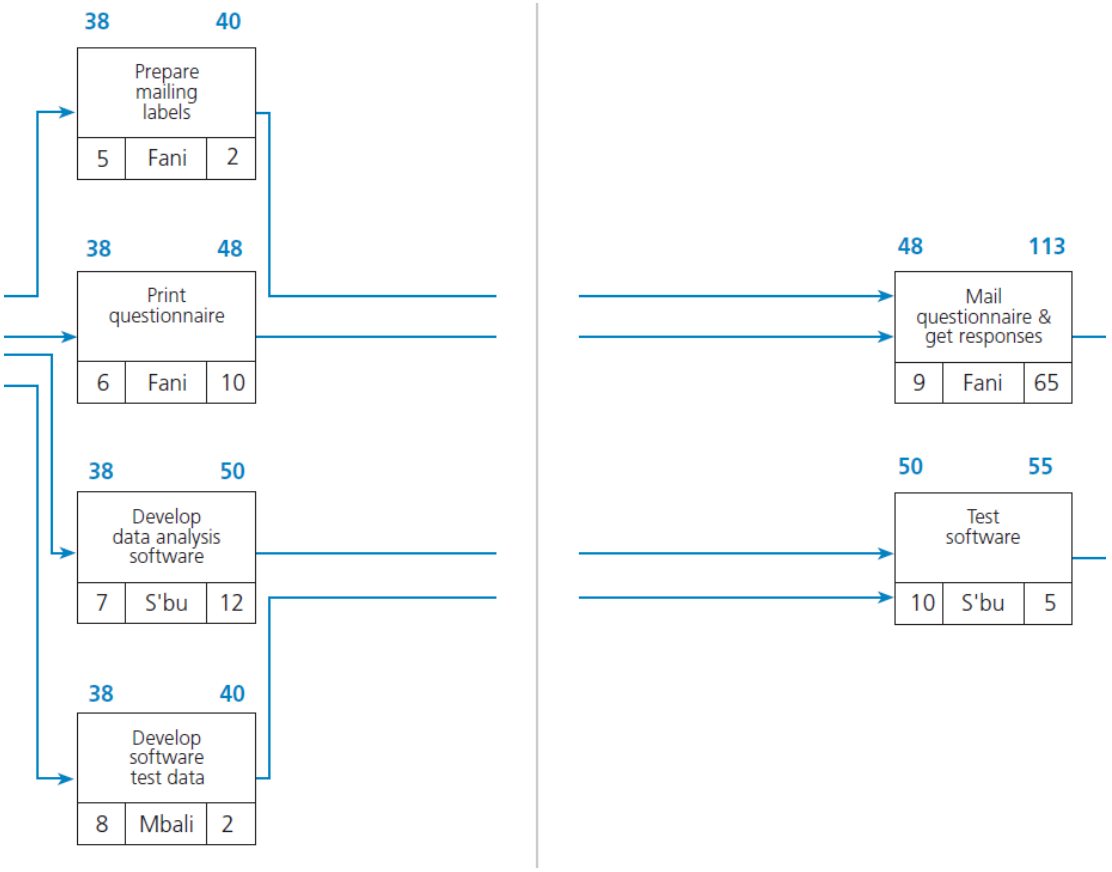
- ES = EF Task 4 = 38
- Task 5
  - Duration = 2
  - EF = 38 + 2 = 40
- Task 6
  - Duration = 10
  - EF = 38 + 10 = 48
- Task 7
  - Duration = 12
  - EF = 38 + 12 = 50
- Task 8
  - Duration = 2
  - EF = 38 + 2 = 40



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'Mail questionnaire & get responses'

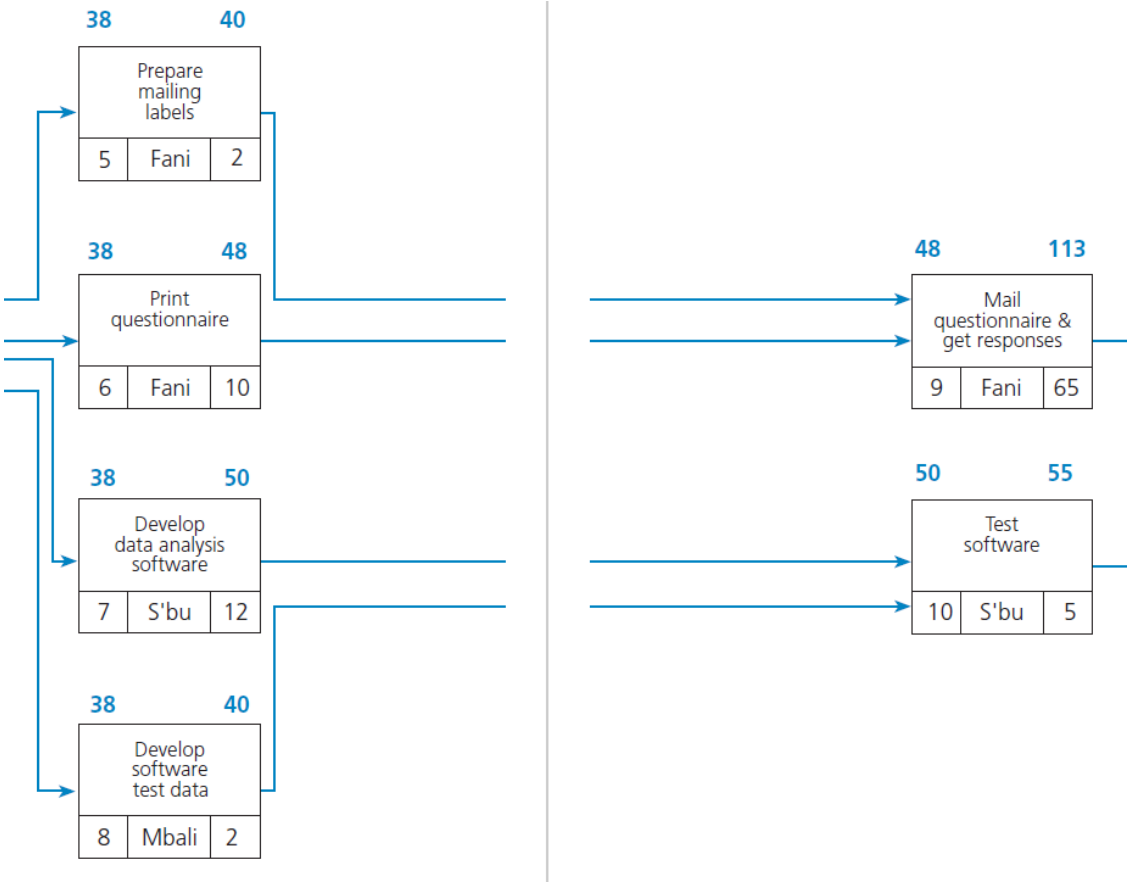
- Latest of Tasks 5 and 6 = 48
- ES = EF Task 6 = 48
- Duration = 65
- EF = 48 + 65 = 113



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'Test software'

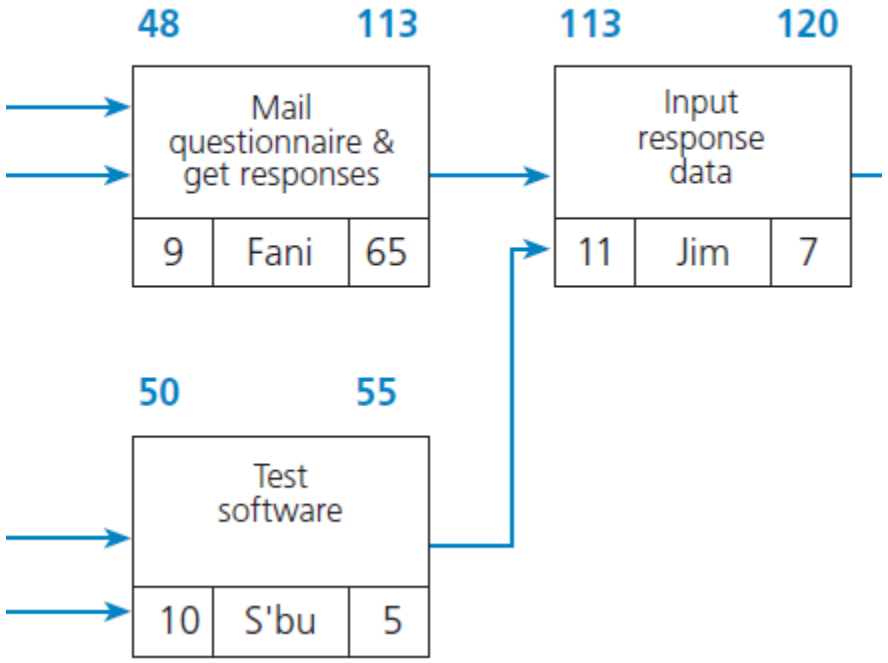
- Latest of Tasks 7 and 8 = 50
- ES = EF Task 7 = 50
- Duration = 5
- EF = 50 + 5 = 55



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'Input response data'

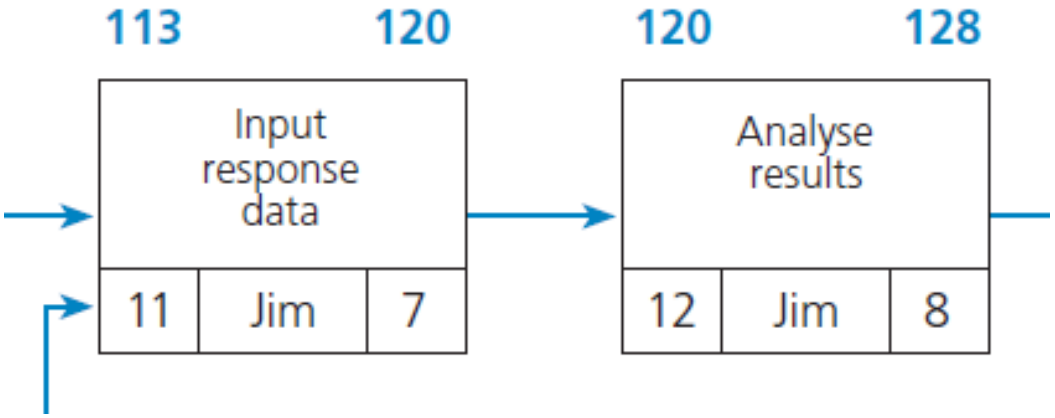
- Latest of Tasks 9 and 10 = 113
- ES = EF Task 9 = 113
- Duration = 7
- EF = 113 + 7 = 120



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'Analyse results'

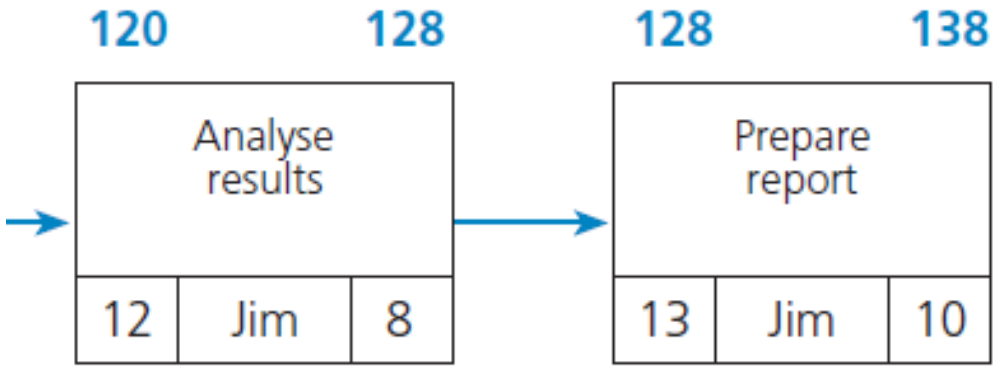
- ES = EF Task 11 = 120
- Duration = 8
- EF = 120 + 8 = 128



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'Prepare report'

- ES = EF Task 12 = 128
- Duration = 10
- EF = 128 + 10 = 138



- Project not complete in required time

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Schedule table ES and EF

FIGURE 5.5 Schedule for Consumer Market Study Project, Showing Earliest Start and Finish Times

Consumer market study project			Estim. Dur.	Earliest					
	Activity	Respon.		Start	Finish				
1	Identify target consumers	Mbali	3	**0**	**3**				
2	Develop draft questionnaire	Mbali	10	**3**	**13**				
3	Pilot-test questionnaire	Mbali	20	**13**	**33**				
4	Review comments & finalise questionnaire	Mbali	5	**33**	**38**				
5	Prepare mailing labels	Fani	2	**38**	**40**				
6	Print questionnaire	Fani	10	**38**	**48**				
7	Develop data analysis software	S'bu	12	**38**	**50**				
8	Develop software test data	Mbali	2	**38**	**40**				
9	Mail questionnaire & get responses	Fani	65	**48**	**113**				
10	Test software	S'bu	5	**50**	**55**				
11	Input response data	Jim	7	**113**	**120**				
12	Analyse results	Jim	8	**120**	**128**				
13	Prepare report	Jim	10	**128**	**138**				

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B. Latest start and finish times

- Latest start time (LS)
  - Latest time by which a specific activity must be **started**
- Latest finish time (LF)
  - Latest time by which a specific activity must be **completed**
- LS = LF – Estimated duration
- The LF and LS times are determined by calculating *backward* through the network diagram from the end of the project to the beginning of the project
- One rule that must be followed in making these backward calculations
- ***Rule 2:*** *The latest finish time for a specific activity must be the same as or earlier than the earliest of all the latest start times of all the activities that flow directly from that specific activity*

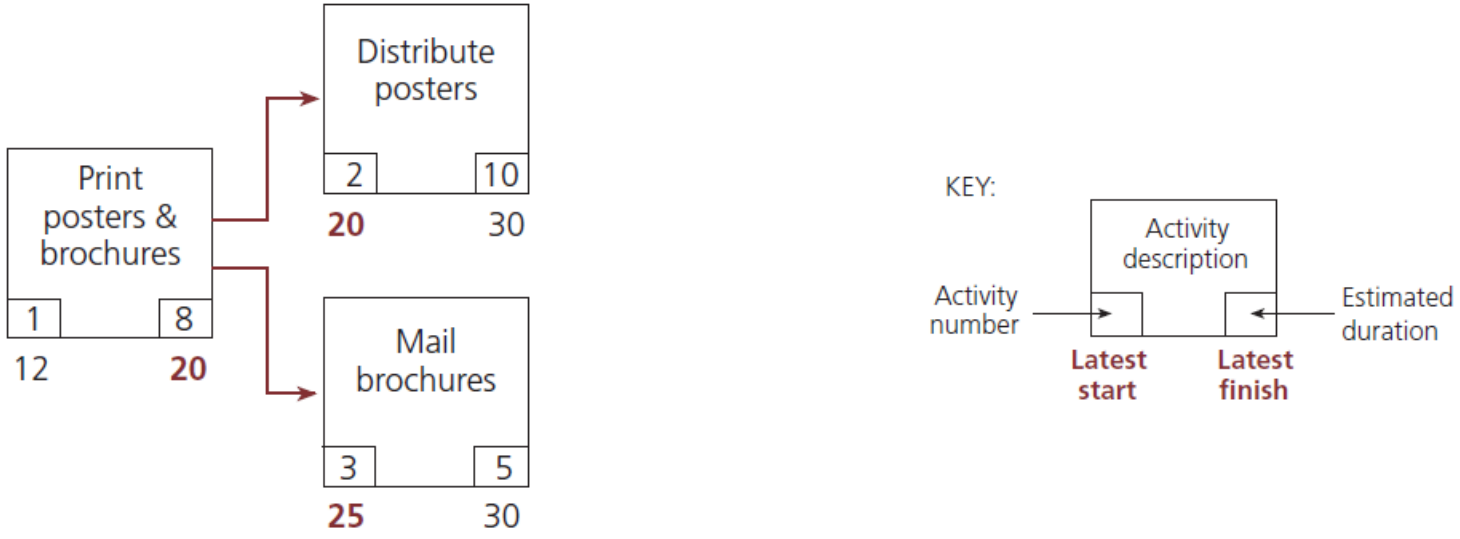
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Figure 5.6 (page 159): Latest Finish Times

Latest start and finish times calculation

- Why is the LF for 'Print posters & brochures' 20?

FIGURE 5.6 Latest Finish Times



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C. Total slack

- Sometimes called *float*
- The difference between EF time of last activity and the project required completion time
- Negative slack
  - Lack of slack over the entire project
  - Amount of time an activity must be accelerated
- Positive slack
  - Maximum amount of time that the activities on a particular path can be delayed without jeopardising on-time completion

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- *The total slack for a particular path of activities is common to and shared among all the activities on that path*
- **Example:** unnumbered image on pg 161 depicts a project to remove old wallpaper and put-up new wallpaper. The total completion time is 20 days. The tasks take 15 days (7 + 5 +3). The project has 5 days of slack

$$Total\ Slack = LF - EF$$

Or

$$Total\ Slack = LS - ES$$

- The two calculations are equivalent

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D. Critical path

- Longest path in the overall network diagram
- Find which activities have the least amount of slack

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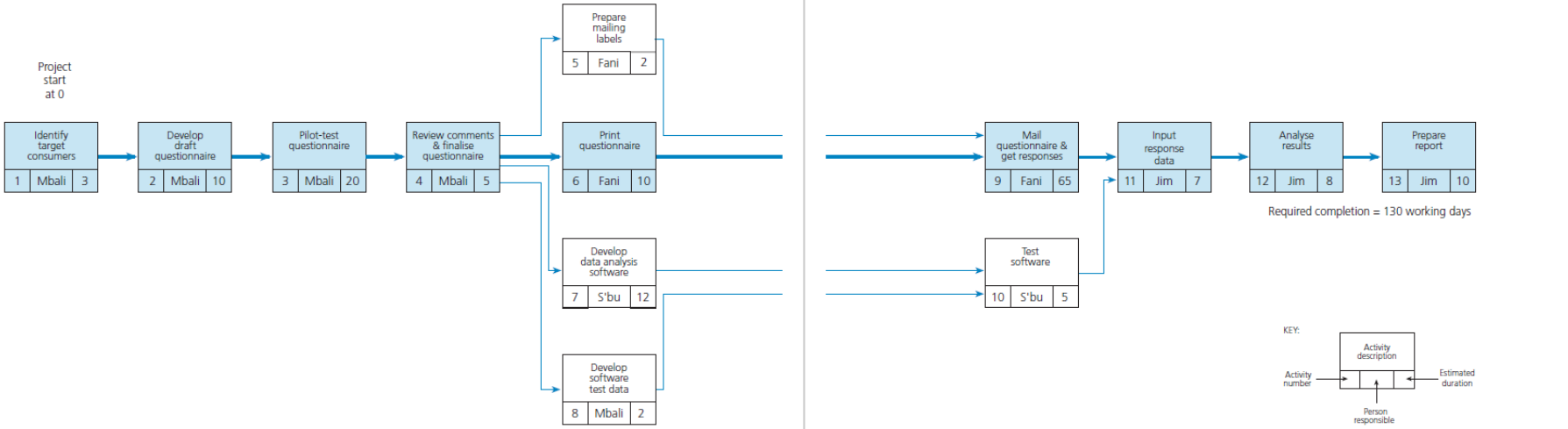
Critical path through a project

FIGURE 5.9 Schedule for Consumer Market Study Project, Showing Total Slack Values

Consumer market study project

	Activity	Respon.	Estim. Dur.	Earliest		Latest		Total slack	
				Start	Finish	Start	Finish		
1	Identify target consumers	Mbali	3	0	3	–8	–5	–8	
2	Develop draft questionnaire	Mbali	10	3	13	–5	5	–8	
3	Pilot-test questionnaire	Mbali	20	13	33	5	25	–8	
4	Review comments & finalise questionnaire	Mbali	5	33	38	25	30	–8	
5	Prepare mailing labels	Fani	2	38	40	38	40	0	
6	Print questionnaire	Fani	10	38	48	30	40	–8	
7	Develop data analysis software	S'bu	12	38	50	88	100	50	
8	Develop software test data	Mbali	2	38	40	98	100	60	
9	Mail questionnaire & get responses	Fani	65	48	113	40	105	–8	
10	Test software	S'bu	5	50	55	100	105	50	
11	Input response data	Jim	7	113	120	105	112	–8	
12	Analyse results	Jim	8	120	128	112	120	–8	
13	Prepare report	Jim	10	128	138	120	130	–8	

FIGURE 5.10 Network Diagram for Consumer Market Study Project, Showing the Critical Path



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Change in slack for critical path

FIGURE 5.11 Revised Schedule for Consumer Market Study Project

Consumer market study project			Estim. dur.	Earliest		Latest		Total slack	
Activity		Respon.		Start	Finish	Start	Finish		
1	Identify target consumers	Mbali	3	0	3	2	5	2	
2	Develop draft questionnaire	Mbali	10	3	13	5	15	2	
3	Pilot-test questionnaire	Mbali	20	13	33	15	35	2	
4	Review comments & finalise questionnaire	Mbali	5	33	38	35	40	2	
5	Prepare mailing labels	Fani	2	38	40	48	50	10	
6	Print questionnaire	Fani	10	38	48	40	50	2	
7	Develop data analysis software	S'bu	12	38	50	88	100	50	
8	Develop software test data	Mbali	2	38	40	98	100	60	
9	Mail questionnaire & get responses	Fani	55	48	103	50	105	2	
10	Test software	S'bu	5	50	55	100	105	50	
11	Input response data	Jim	7	103	110	105	112	2	
12	Analyse results	Jim	8	110	118	112	120	2	
13	Prepare report	Jim	10	118	128	120	130	2	

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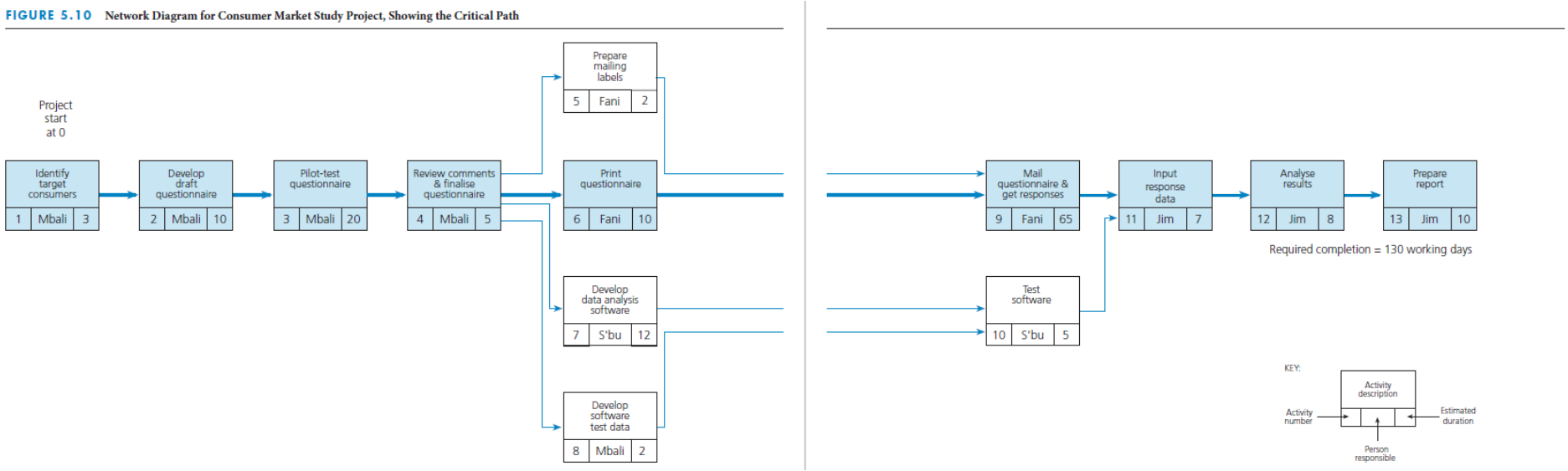
E. Free slack

- Time a specific activity can be postponed without delaying the ES of its immediate successor activities
- Calculation
  - Find lowest of the values of total slack for all the activities entering into a specific activity
  - Subtract value from the values of total slack for the other activities also entering into that same activity
- Free slack will exist only when two or more activities enter into the same activity
- FS is always a positive value

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Total slack compared to free slack

- Total slack for Activity 7 = 50
- Total slack for Activity 8 = 60
- Free slack for Activity 8 = 60 – 50 = 10 days

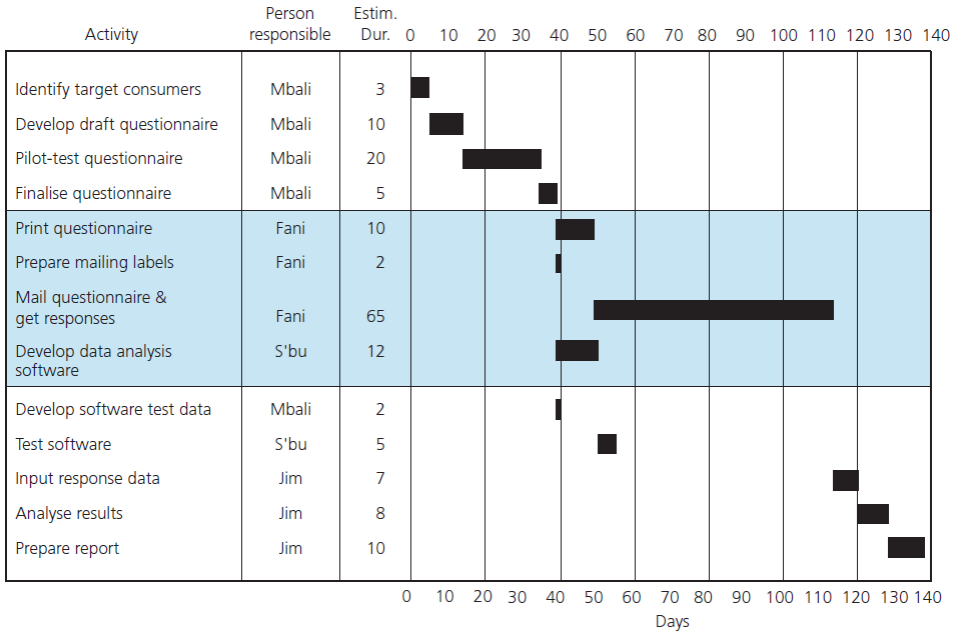


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F. Bar chart format

- Gantt chart tool for planning and scheduling
- Activities on side
- Time scale on top or bottom
- Estimated duration in bars
- Automatically generated in software systems
- Can show relationships between activities

FIGURE 5.12 Bar Chart for Consumer Market Study Project

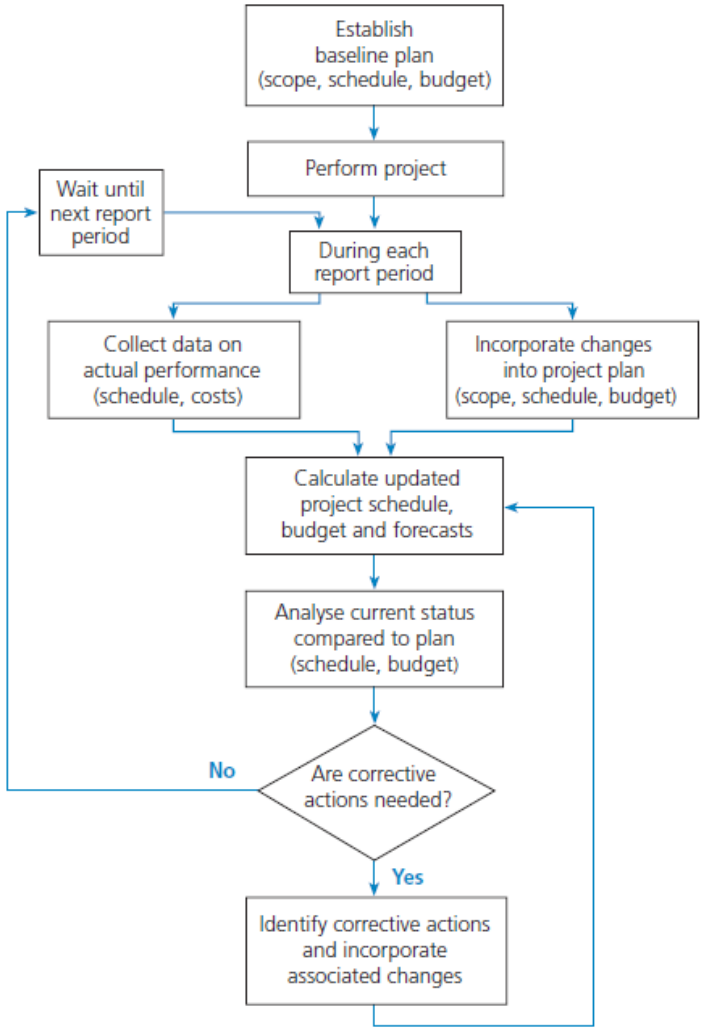


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Project Control Process

- Meetings occur regularly
- Gather data on actual performance
- Record changes
- Monitor progress

FIGURE 5.13 Project Control Process

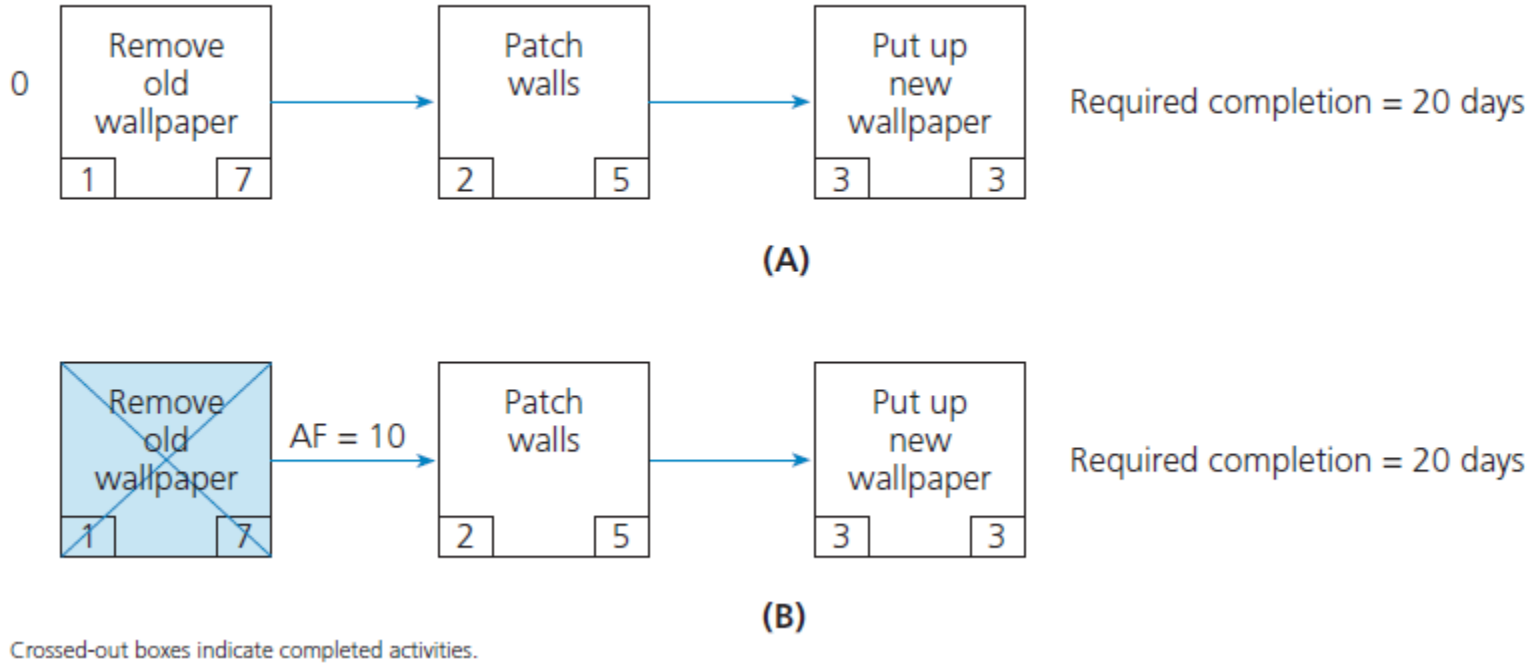


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Effects of Actual Schedule Performance

- Part (a) Total slack = +5

FIGURE 5.14 Effect of Actual Finish Times



Crossed-out boxes indicate completed activities.

- Part (b) Total slack = +2

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Incorporate Changes into Schedule

- Changes may impact the schedule
  - Initiated by customer or project team
  - Result from unanticipated occurrence
- Early change may have less impact than later change
- Manage requested changes
  - Estimate impact
  - Obtain customer approval
  - Revise project plan, schedule, and costs

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Update Project Schedule

- Generate forecasts for project finish
  - Use actual finish dates of completed activities
  - Enter project changes
  - Update project schedule
- Determine if any changes occur in critical path

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Control Schedule

Schedule control steps

1. Analyse the schedule for needed corrective action
2. Decide specific corrective actions to be taken
3. Revise the plan to incorporate corrective actions
4. Recalculate the schedule to evaluate the effects of the planned corrective actions

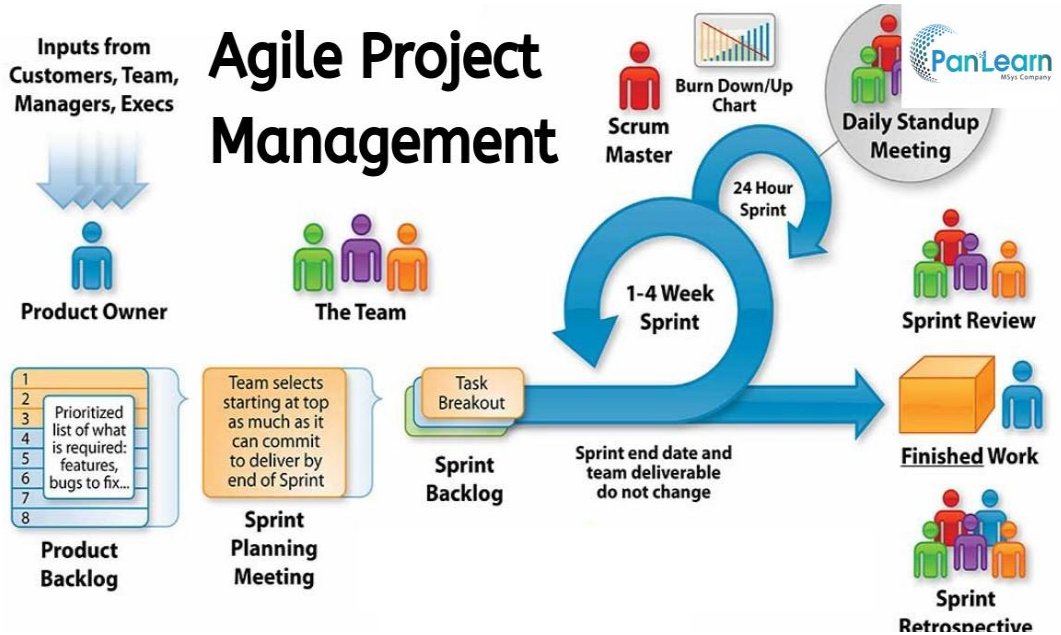
Actions

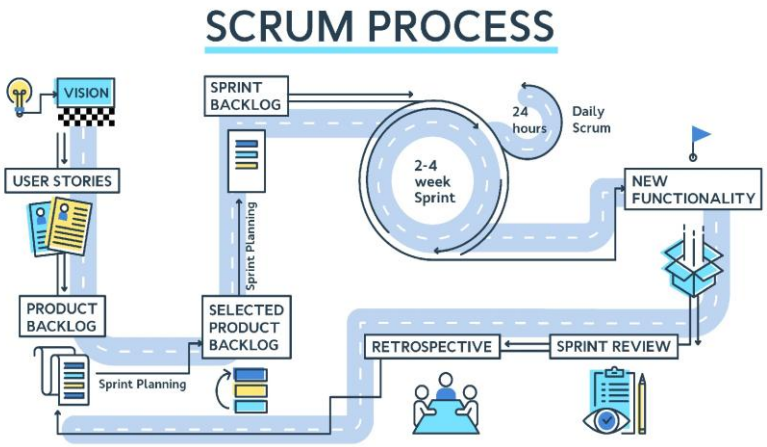
- Repeat steps if not acceptable results
- Apply efforts to paths with negative slack
  - Near-term activities
  - Long estimated durations
- Change may shift critical path
- Trade-off of costs and scope

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Scheduling for Information Systems Development and Project Management Information Systems

- Individual Reading
  - pages 180 till 192





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IS example: activities, predecessors, durations

FIGURE 5.17 List of Activities, Immediate Predecessors and Estimated Durations Web-Based Reporting System Project

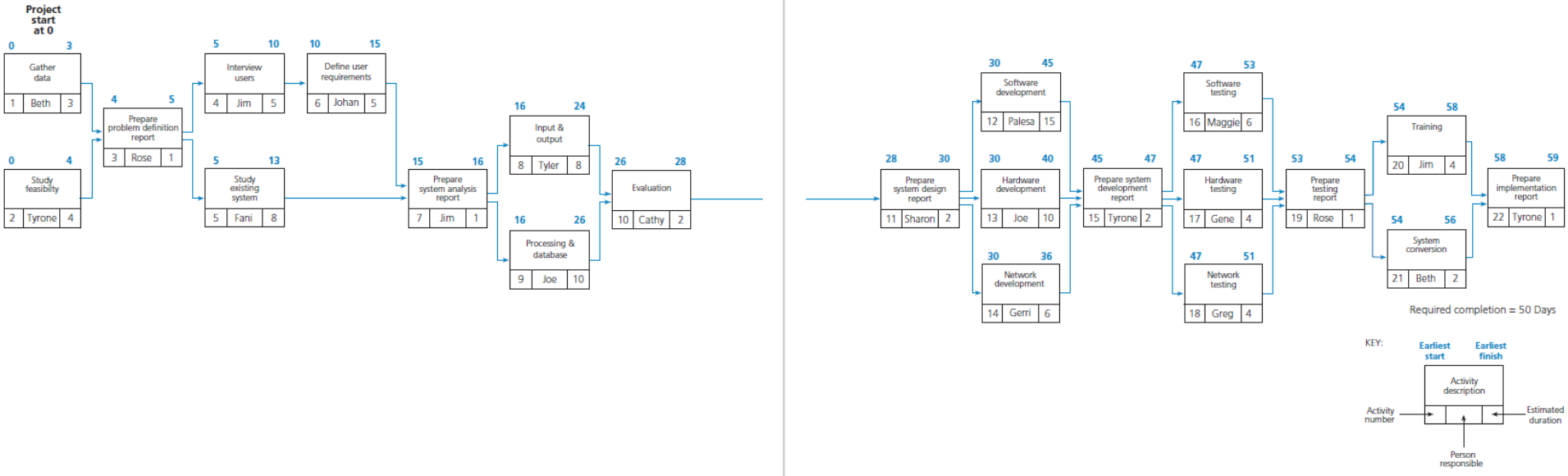
Web-based reporting system project

Activity	Immediate predecessors	Estimated duration (days)
1. Gather data	—	3
2. Study feasibility	—	4
3. Prepare problem definition report	1, 2	1
4. Interview users	3	5
5. Study existing system	3	8
6. Define user requirements	4	5
7. Prepare system analysis report	5, 6	1
8. Input & output	7	8
9. Processing & database	7	10
10. Evaluation	8, 9	2
11. Prepare system design report	10	2
12. Software development	11	15
13. Hardware development	11	10
14. Network development	11	6
15. Prepare system development report	12, 13, 14	2
16. Software testing	15	6
17. Hardware testing	15	4
18. Network testing	15	4
19. Prepare testing report	16, 17, 18	1
20. Training	19	4
21. System conversion	19	2
22. Prepare implementation report	20, 21	1

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IS example: ES and EF times

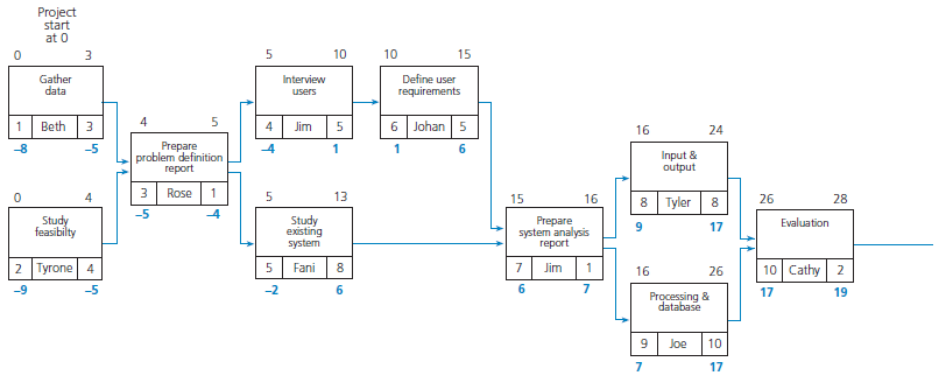
FIGURE 5.18 Network Diagram for Web-Based Reporting System Project, Showing Earliest Start and Finish Times

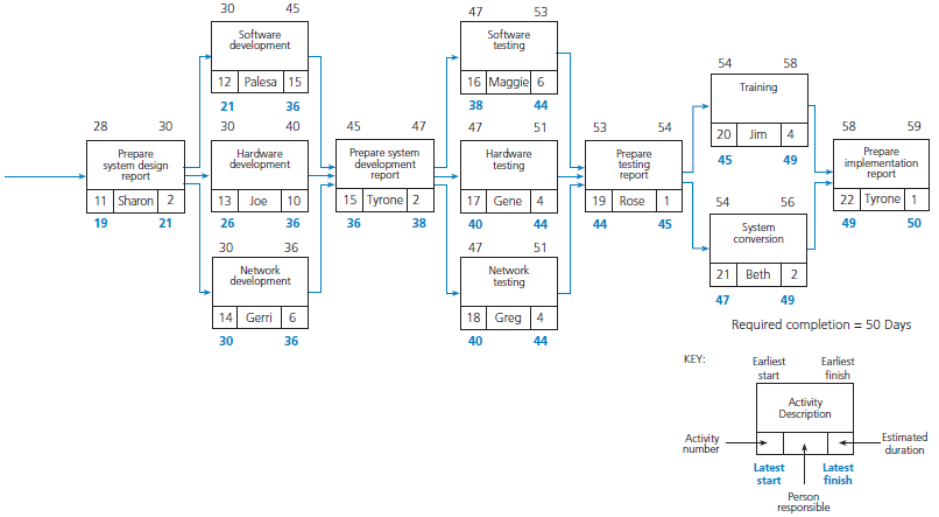


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IS example: LS and LF times

FIGURE 5.19 Network Diagram for Web-Based Reporting System Project, Showing Latest Start and Finish Times





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IS example: schedule table

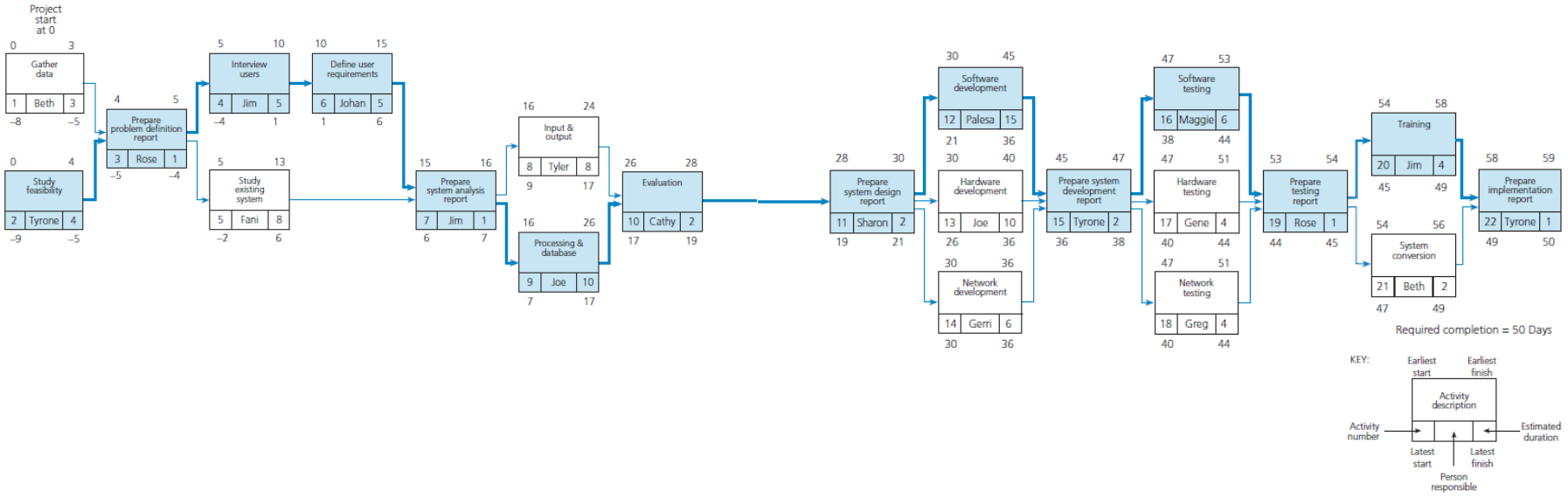
FIGURE 5.20 Schedule for Web-Based Reporting System Project

Web-based reporting system project			Estim. Dur.	Earliest		Latest		Total slack	
	Activity	Respon.		Start	Finish	Start	Finish		
1	Gather data	Beth	3	0	3	–8	–5	–8	
2	Study feasibility	Tyrone	4	0	4	–9	–5	–9	
3	Prepare problem definition report	Rose	1	4	5	–5	–4	–9	
4	Interview users	Jim	5	5	10	–4	1	–9	
5	Study existing system	Fani	8	5	13	–2	6	–7	
6	Define user requirements	Johan	5	10	15	1	6	–9	
7	Prepare system analysis report	Jim	1	15	16	6	7	–9	
8	Input & output	Tyler	8	16	24	9	17	–7	
9	Processing & database	Joe	10	16	26	7	17	–9	
10	Evaluation	Cathy	2	26	28	17	19	–9	
11	Prepare system design report	Sharon	2	28	30	19	21	–9	
12	Software development	Palesa	15	30	45	21	36	–9	
13	Hardware development	Joe	10	30	40	26	36	–4	
14	Network development	Gerri	6	30	36	30	36	0	
15	Prepare system development report	Tyrone	2	45	47	36	38	–9	
16	Software testing	Maggie	6	47	53	38	44	–9	
17	Hardware testing	Gene	4	47	51	40	44	–7	
18	Network testing	Greg	4	47	51	40	44	–7	
19	Prepare testing report	Rose	1	53	54	44	45	–9	
20	Training	Jim	4	54	58	45	49	–9	
21	System conversion	Beth	2	54	56	47	49	–7	
22	Prepare implementation report	Tyrone	1	58	59	49	50	–9	

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IS example: critical path

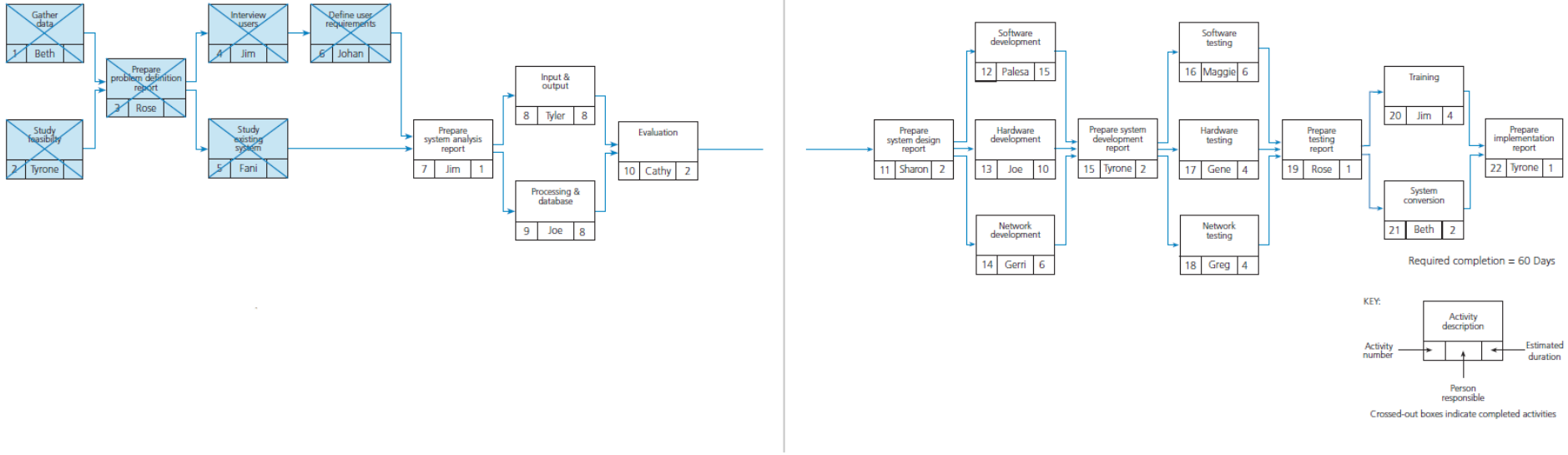
FIGURE 5.21 Network Diagram for Web-Based Reporting System Project, Showing the Critical Path



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IS example: updated network diagram

FIGURE 5.22 Network Diagram for Web-Based Reporting System Project Incorporating Actual Progress and Changes



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IS example: updated schedule table

FIGURE 5.23 Updated Schedule for Web-Based Reporting System Project

Web-based reporting system project			Estim. dur.	Earliest		Latest		Total slack	Actual finish
	Activity	Respon.		Start	Finish	Start	Finish		
1	Gather data	Beth							4
2	Study feasibility	Tyrone							4
3	Prepare problem definition report	Rose							5
4	Interview users	Jim							10
5	Study existing system	Fani							15
6	Define user requirements	Johan							18
7	Prepare system analysis report	Jim	1	18	19	18	19	0	
8	Input & output	Tyler	8	19	27	19	27	0	
9	Processing & database	Joe	8	19	27	19	27	0	
10	Evaluation	Cathy	2	27	29	27	29	0	
11	Prepare system design report	Sharon	2	29	31	29	31	0	
12	Software development	Palesa	15	31	46	31	46	0	
13	Hardware development	Joe	10	31	41	36	46	5	
14	Network development	Gerri	6	31	37	40	46	9	
15	Prepare system development report	Tyrone	2	46	48	46	48	0	
16	Software testing	Maggie	6	48	54	48	54	0	
17	Hardware testing	Gene	4	48	52	50	54	2	
18	Network testing	Greg	4	48	52	50	54	2	
19	Prepare testing report	Rose	1	54	55	54	55	0	
20	Training	Jim	4	55	59	55	59	0	
21	System conversion	Beth	2	55	57	57	59	2	
22	Prepare implementation report	Tyrone	1	59	60	59	60	0	

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Project Management Information Systems

- Most systems perform scheduling functions
- Calculates at click of the mouse
  - ES, EF, LS, and LF
  - Total slack
  - Critical path
- Perform control functions

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Agile Project Management

Scrum approach participants

- Product owner
- Development team
- Scrum master

Process includes

1. Establish and authorise
2. Define requirements
3. Conduct daily scrum meetings
4. Conduct sprint review meeting
5. Conduct sprint retrospective meeting

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Critical Success Factors

- The **person** who will be **responsible** for performing the activity **should estimate** the duration for that activity. This generates commitment from the person.
- The estimated duration for an activity must be **based on the types and quantities** of resources required to perform the activity.
- Activity estimated durations should be **aggressive yet realistic**.
- Activities should not be longer in estimated duration than the **time intervals** at which the actual progress will be reviewed and compared to planned progress.
- Project management involves a **proactive approach** to controlling a project to ensure that the project objective is accomplished even when things do not go according to plan.
- Once the project starts, it is important to **monitor progress** to ensure that everything is going according to plan.
- The key to effective project control is **measuring actual progress and comparing it to planned progress** on a timely and regular basis and taking any needed corrective action immediately.
- The key to effective schedule control is to address any paths with **negative or deteriorating slack values aggressively** as soon as they are identified. A concentrated effort to accelerate project progress must be applied to these paths.

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Critical Success Factors (continued)

- The amount of negative slack should **determine the priority** for applying these concentrated efforts.
- When attempting to reduce the duration of a path of activities that has negative slack**, focus on** activities that are **near term** and on activities that have **long estimated durations**.
- **Addressing schedule problems early** will minimise the negative impact on scope and budget.
- If a project falls too far behind, **getting it back on schedule** becomes more difficult, and usually requires spending more money or reducing the scope or quality.
- If corrective actions are necessary, decisions must be made regarding a **trade-off of scope, time, and cost**.
- A **regular reporting period should be established** for comparing actual progress to planned progress.
- The shorter the reporting period, the **better the chances** of identifying problems early and taking corrective actions.
- During each reporting period, data on **actual performance** and information on **changes** to the project scope, schedule, and budget need to be collected in a timely manner and used to calculate an updated schedule and budget.

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Summary

- The scheduling function depends on the planning function.
- The estimated types and quantities of resources required for an activity, together with the availability of those resources, will influence the estimated duration for how long it will take to perform the activity.
- The estimated duration for each activity must be the total elapsed time – the time for the work to be done plus any associated waiting time.
- The estimate should be aggressive yet realistic.
- It may be easier to estimate the durations for near-term activities, but as the project progresses, the project team can progressively elaborate the estimated the durations as more information becomes known to allow for more accurate estimated durations.
- A project schedule provides a timetable for each activity and shows the earliest start (ES) and earliest finish (EF) times and the latest start (LS) and latest finish (LF) times for each activity.
- The total slack for a particular path of activities through the network is common to and shared among all activities on that path.

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Summary (continued)

- The critical path is the longest (most time-consuming) path of activities in the network diagram.
- The key to effective project control is measuring actual progress and comparing it to planned progress on a timely and regular basis and taking any needed corrective action immediately.
- Actual progress – whether faster or slower than planned – will have an effect on the schedule of the remaining, incomplete activities of the project.
- Any type of change – whether initiated by the customer, the contractor, the project manager, a team member, or an unanticipated event – will require a modification to the plan in terms of scope, schedule, and/or budget.
- Schedule control involves four steps: analysing the schedule to determine which areas may need corrective action, deciding what specific corrective actions should be taken, revising the plan to incorporate the chosen corrective actions, and recalculating the schedule to evaluate the effects of the planned corrective actions.
- One of the most important factors in effective scheduling is estimating activity durations that are as realistic as possible.

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Self Study: Chapter 6 (pages 221 to 237)

- Taking resource constraints into account when developing a network diagram
- Determining the resource requirements plan for a project
- levelling the use of resources within the required time frame of the project
- Determining the shortest project schedule with the limited resources available

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